JOHNS HOPKINS
WHITING SCHOOL of ENGINEERING

FPGA Design Using VHDL

Module 2F: Latches

Latching (Unintentional)

- Latching in the `if/then` syntax occurs when the conditional does not end with an `else`

```
show_latch : process(pushButtons)
begin
   if(pushButtons = "01") then
      an_i <= "0010";
   elsif(pushButtons = "10") then
      an_i <= "0011";
   elsif(pushButtons = "11") then
      an_i <= "1111";
   end if;
end process;
```

- When this is synthesized:
  - `WARNING: [Synth 8-327] inferring latch for variable 'an_i_reg' [D:/Projects/FPGA/Module2F_Latches.srcs/lecture2e_top.vhd:46]`

- Utilization Report:

```
| Slice Registers          |    2 |      0 |           0 |    126800 | <0.01 |
|   Register as Flip Flop  |    0 |      0 |           0 |    126800 |  0.00 |
|   Register as Latch      |    2 |      0 |           0 |    126800 | <0.01 |
```

Latch Avoidance

- Avoid latching when you don't want it

```
show_noLatch : process(pushButtons)
begin
   if(pushButtons = "01") then
      an_i <= "0010";
   elsif(pushButtons = "10") then
      an_i <= "0011";
   elsif(pushButtons = "11") then
      an_i <= "1111";
   else
      an_i <= "0001";
   end if;
end process;
```

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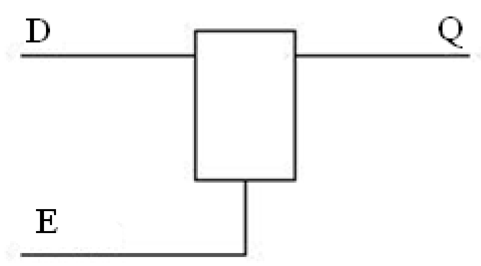
Latch Inference - Example

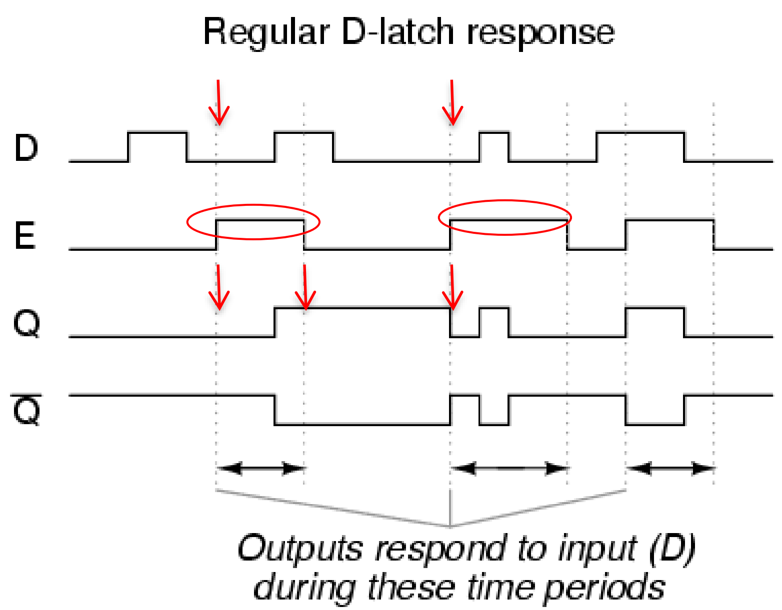
- Synthesizer detects signals which are to be latched (those that are not assigned new value under every condition)
- For simplicity, clearly specify latching when desired (enable)

```
process(en, sel, a, b)
begin
   if(en = '1') then
      if sel='0' then
         z <= a;
      else
         z <= b;
      end if;
   end if;
end process;
```

Latch Inference - Operation

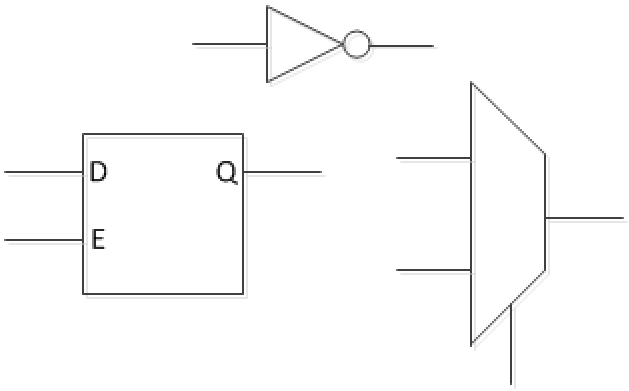



```
process(D, E)
begin
    if(E = '1') then
        Q <= D;
    end if;
end process;
```



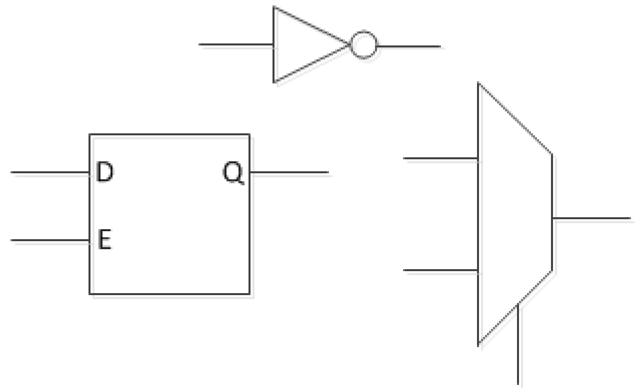
Illustrating Latch Inference - Example

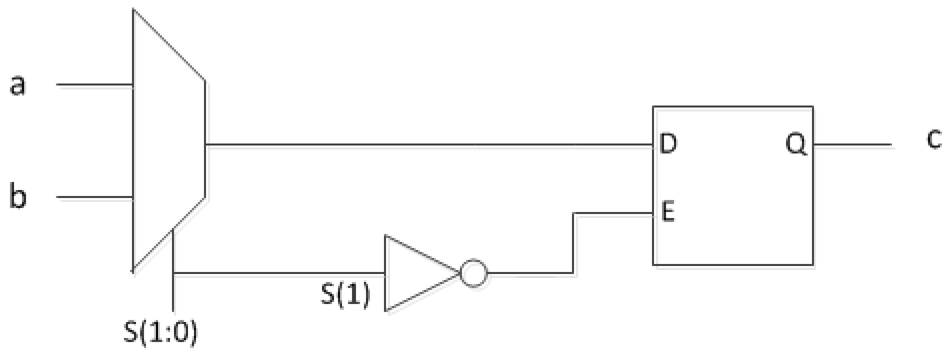
```
process(s, a, b)
begin
    if(s = "00") then
        c <= a;
    elsif(s = "01") then
        c <= b;
    end if;
end process;
```



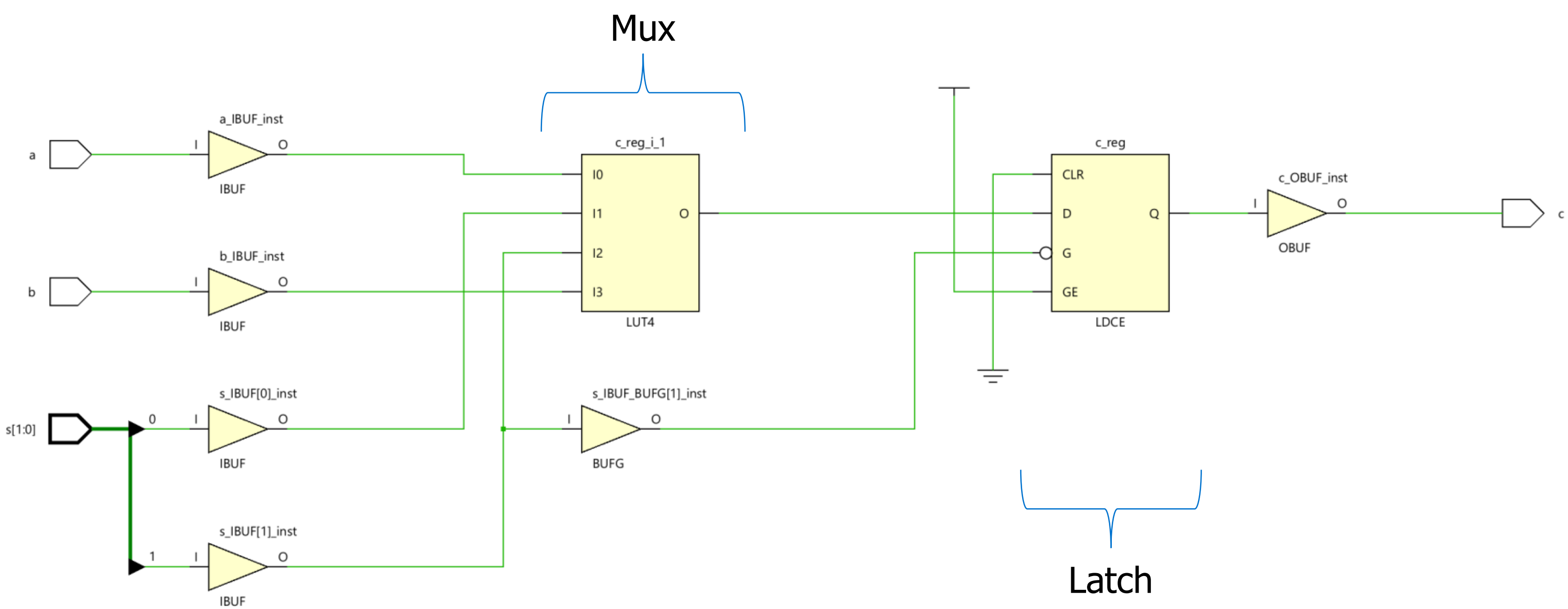
Illustrating Latch Inference - Component

```
process(s, a, b)
begin
    if(s = "00") then
        c <= a;
    elsif(s = "01") then
        c <= b;
    end if;
end process;
```





Illustrating Latch Inference - Schematic



Synthesis Results for Latch

```
LIBRARY ieee;
USE ieee.std_logic_1164.ALL;
ENTITY Latchvsnolatch IS port (
   a, b : in std_logic
   s : in std_logic_vector(1 downto 0);
   c: out std_logic);
END Latchvsnolatch;

ARCHITECTURE arch OF Latchvsnolatch IS

BEGIN

process(s, a, b)
begin
   if(s = "00") then
      c <= a;
   elsif(s = "01") then
      c <= b;
   end if;
end process;
END arch ;
```

Site Type	Used	Fixed	Prohibited	Available	Util%
Slice LUTs*	1	0	0	63400	<0.01
LUT as Logic	1	0	0	63400	<0.01
LUT as Memory	0	0	0	19000	0.00
Slice Registers	1	0	0	126800	<0.01
Register as Flip Flop	0	0	0	126800	0.00
Register as Latch	1	0	0	126800	<0.01
F7 Muxes	0	0	0	31700	0.00
F8 Muxes	0	0	0	15850	0.00